

# Exactness of ARIEL on Singly-Connected LCNs

## Chains, Trees, Polytrees — and How the Factor Graph Preserves the Local Markov Condition

IBM Research

### Abstract

ARIEL (Marinescu et al., *Approximate Inference in LCNs*, IJCAI-2023) is a belief-propagation-style message-passing scheme for marginal inference in Logical Credal Networks (LCNs). Its central guarantee, **Theorem 1**, states that on a *singly-connected* LCN it returns the *exact* posterior marginal bounds for every atom. This note re-examines that claim against the implementation (`lcn/inference/marginal/ariel.py`) on three singly-connected structures of increasing generality — a Markov **chain**, a branching **tree**, and a **polytree** (with colliders) — using an independent, solver-robust ground-truth oracle, and confirms ARIEL is exact on all three to four decimals. The heart of the note is a careful analysis of *how* the Local Markov Condition (LMC) independencies are preserved. The perhaps-surprising finding is that on these structures *not a single LMC assertion fits inside one factor*, so *none is imposed as an explicit constraint*; instead the LMC is preserved *structurally*, by the bipartite factor *tree* together with the interval-message protocol. We make this precise, draw the tree’s factor graph, and identify the exact boundary: on a *loopy* factor graph the interval messages can no longer carry the joint information a spanning LMC (or a cross-family sentence) needs, and ARIEL degrades to a sound but non-tight outer bound.

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## 1 Background: the ARIEL algorithm

### 1.1 LCNs and the marginal task

An LCN over binary atoms  $\mathbf{V} = \{V_1, \dots, V_n\}$  is a set of probability *sentences*, each either Type-1,  $\ell \leq P(\varphi) \leq u$ , or Type-2,  $\ell \leq P(\varphi \mid \psi) \leq u$ , over propositional formulas  $\varphi, \psi$ . The sentences, together with the conditional independencies derived from the *Local Markov Condition* (LMC) of the LCN’s primal graph, define a set  $\mathcal{D}_{\text{LCN}} \subseteq \Delta(2^{\mathbf{V}})$  of joint distributions. The *marginal task* for a query atom  $a$  is to compute

$$\underline{P}(a) = \min_{p \in \mathcal{D}_{\text{LCN}}} P(a), \quad \overline{P}(a) = \max_{p \in \mathcal{D}_{\text{LCN}}} P(a).$$

Exact inference solves one nonconvex NLP over the full  $2^n$  joint per bound; this is what **ExactInference** (`exact.py`) does, and it is intractable beyond  $\sim 10$  atoms.

### 1.2 The Local Markov Condition

Given the LCN’s primal graph  $G$ , the LMC states that every atom  $x$  is conditionally independent of its non-descendant, non-parent atoms given its parents. Concretely, for each atom  $x$  with parent set  $S = \text{pa}(x)$  and non-descendant non-parents  $Y$ , the LMC contributes the assertion  $(x \perp\!\!\!\perp Y \mid S)$ . For binary atoms this is encoded (as in `exact.py` and `lmc_constraint_groups_vec`) by the bilinear *joint factorization*

$$P(x, y, s) P(s) = P(x, s) P(y, s) \quad \text{for every joint configuration of } Y \text{ and } S, \quad (1)$$

which is a set of quadratic equalities over the joint  $\vec{p}$ . These equalities are part of the definition of  $\mathcal{D}_{\text{LCN}}$  — dropping them *enlarges* the feasible set and can only loosen the bounds. The central question of Section 5 is therefore: does ARIEL preserve (1)?

### 1.3 The factor graph

ARIEL works on a *factor graph*  $\mathcal{F}$ : a bipartite graph of *variable nodes* (the atoms) and *factor nodes*. A factor node groups all sentences that share the *same* atom scope, and is connected to each variable node in that scope. For example, the two transition sentences  $P(x_2 \mid x_1)$  and  $P(x_2 \mid \neg x_1)$  of a chain share scope  $\{x_1, x_2\}$  and form one factor; a prior  $P(x_0)$  forms a singleton factor  $\{x_0\}$ .

### 1.4 Message passing

ARIEL passes *interval* messages along the edges and tightens them iteratively (Algorithm 1). There are two kinds.

**Variable-to-factor**  $v \rightarrow f$ . The intersection of all *other* factor-to-variable intervals incident on  $v$ :

$$\underline{\ell}_{v \rightarrow f} = \max_{f' \in N(v) \setminus \{f\}} \underline{\ell}_{f' \rightarrow v}, \quad \overline{u}_{v \rightarrow f} = \min_{f' \in N(v) \setminus \{f\}} \overline{u}_{f' \rightarrow v}.$$

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**Algorithm 1** ARIEL (Marinescu et al., IJCAI-2023, Algorithm 1)

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**Require:** LCN  $\mathcal{L}$ ; factor graph  $\mathcal{F}$

- 1: initialize every edge message to  $[0, 1]$
  - 2: **repeat**
  - 3:   **for all** edges  $(v, f)$  **do** update  $v \rightarrow f$  by intersecting the other  $f' \rightarrow v$  messages
  - 4:   **end for**
  - 5:   **for all** edges  $(v, f)$  **do** set  $\underline{\ell}_{f \rightarrow v}, \bar{u}_{f \rightarrow v} = \min, \max P(v)$  over the local NLP (sentences + in-scope LMC + neighbour intervals)
  - 6:   **end for**
  - 7: **until** converged
  - 8: **for all** atoms  $v$  **do return**  $\underline{P}(v) = \max_f \underline{\ell}_{f \rightarrow v}, \bar{P}(v) = \min_f \bar{u}_{f \rightarrow v}$
  - 9: **end for**
- 

**Factor-to-variable**  $f \rightarrow n$ . Minimize and maximize  $P(n)$  over a *local* NLP on the factor’s own joint  $\vec{p} \in \Delta(2^{|\text{scope}(f)|})$ , subject to: (i)  $f$ ’s sentences; (ii) the LMC equalities (1) *whose entire scope*  $x \cup Y \cup S$  *fits inside*  $\text{scope}(f)$ ; and (iii) for every other boundary variable  $m \in \text{scope}(f) \setminus \{n\}$ , its incoming interval  $[\underline{\ell}_{m \rightarrow f}, \bar{u}_{m \rightarrow f}]$  applied to the singleton marginal  $P(m)$ .

## 1.5 Theorem 1 and the key approximation

The paper’s Theorem 1 asserts ARIEL is *exact* on singly-connected LCNs (the primal graph has no directed cycle), even with extra marginal sentences that break the unique-assessment assumption. The implicit approximation — exactly why exactness is only claimed for the singly-connected case — is that a factor-to-variable message summarizes the other boundary variables  $m \neq n$  *only through their marginal intervals*  $[\underline{\ell}_{m \rightarrow f}, \bar{u}_{m \rightarrow f}]$ . A message carries no *joint* or even *correlation* information across the factor boundary. On a tree this loses nothing, because the boundary variable separates the two sides; on a loopy graph it can lose tightness. Sections 3–6 make both halves concrete, and Section 5 traces precisely how this interacts with the LMC.

## 2 The three singly-connected test structures

We use three structures of increasing generality, all singly connected, all with  $\leq 8$  atoms so that an independent exact oracle is available. They are shipped as `examples/chain.lcn`, `examples/tree.lcn`, and `examples/polytree.lcn`.

All three have only family-local sentences (a prior on each root and  $P(\text{child} \mid \text{parents})$  on each edge), so the only cross-family constraints are the LMC independencies, and every structure is singly connected. By Theorem 1, ARIEL should be exact on all three.

**Ground truth.** We compare against two independent exact references: **CredalJT** (the junction-tree exact NLP, scheme D5 of the companion note), provably exact on these structures; and an LMC-correct *brute-force* oracle that optimizes over a parametric family which satisfies every LMC equality (1) *by construction* (residual  $\sim 10^{-16}$ ), using SLSQP only — hence immune to the nonconvex-solver unreliability that afflicts **ExactInference/SCIP** on these bilinear systems. The two references agree to four decimals on every atom, so we report one “exact” column.

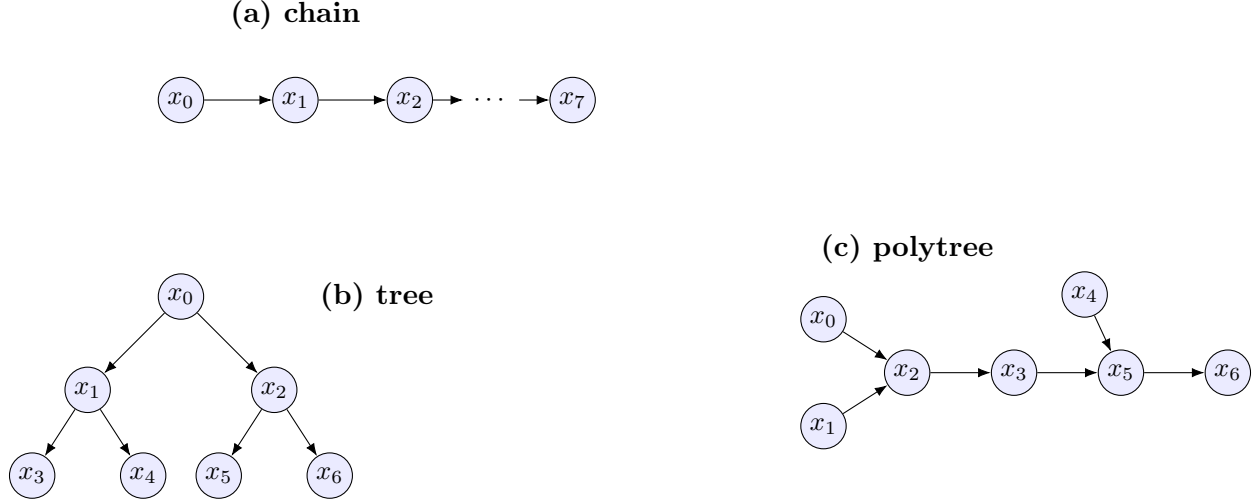


Figure 1: The three singly-connected *primal* graphs. (a) the Markov chain `chain.lcn` ( $n=8$ ); (b) the branching binary tree `tree.lcn` ( $n=7$ , every node has one parent); (c) the polytree `polytree.lcn` ( $n=7$ ) with two colliders  $x_0, x_1 \rightarrow x_2$  and  $x_3, x_4 \rightarrow x_5$  (multi-parent nodes).

### 3 ARIEL is exact on all three structures

Running `ArielInference.run` (`n_iters= 60`, threshold  $10^{-9}$ , fully converged) reproduces the exact marginal bounds on the chain, the tree, and the polytree (Table 1) — to four decimals on every atom, matching both the `CredalJT` and the brute-force oracle. This is exactly what Theorem 1 predicts.

| structure | atom  | ARIEL            | exact            |
|-----------|-------|------------------|------------------|
| chain     | $x_0$ | [0.3000, 0.5000] | [0.3000, 0.5000] |
|           | $x_1$ | [0.4000, 0.6800] | [0.4000, 0.6800] |
|           | $x_2$ | [0.3280, 0.6400] | [0.3280, 0.6400] |
|           | $x_7$ | [0.3336, 0.6667] | [0.3340, 0.6670] |
| tree      | $x_0$ | [0.3000, 0.5000] | [0.3000, 0.5000] |
|           | $x_1$ | [0.4000, 0.6800] | [0.4000, 0.6800] |
|           | $x_3$ | [0.3280, 0.6400] | [0.3280, 0.6400] |
|           | $x_6$ | [0.3280, 0.6400] | [0.3280, 0.6400] |
| polytree  | $x_2$ | [0.3100, 0.6300] | [0.3100, 0.6300] |
|           | $x_3$ | [0.3480, 0.6760] | [0.3480, 0.6760] |
|           | $x_5$ | [0.2644, 0.6528] | [0.2644, 0.6528] |
|           | $x_6$ | [0.3389, 0.6942] | [0.3389, 0.6942] |

Table 1: ARIEL reproduces the exact marginal bounds on all three singly-connected structures (residual differences are solver tolerance,  $\leq 4 \times 10^{-4}$ ), confirming Theorem 1. Bounds are stable from 10 iterations onward.

## 4 The factor graph of the tree, and its bipartite tree shape

To analyse *why* exactness holds we need the factor graph itself, not just the primal graph. For the tree of Figure 1(b), the sentences group by scope into seven factors:

$$f_1=\{x_0, x_1\}, f_2=\{x_0, x_2\}, f_3=\{x_1, x_3\}, f_4=\{x_1, x_4\}, f_5=\{x_2, x_5\}, f_6=\{x_2, x_6\}, f_7=\{x_0\},$$

where  $f_7$  holds the root prior  $P(x_0)$  and each  $f_i$  ( $i \leq 6$ ) holds the two transition sentences  $P(\text{child} \mid \text{parent}), P(\text{child} \mid \neg \text{parent})$  of one tree edge. Figure 2 draws the bipartite factor graph. **It is itself a tree**: there is no cycle through variable and factor nodes. This is the property Theorem 1 hinges on.

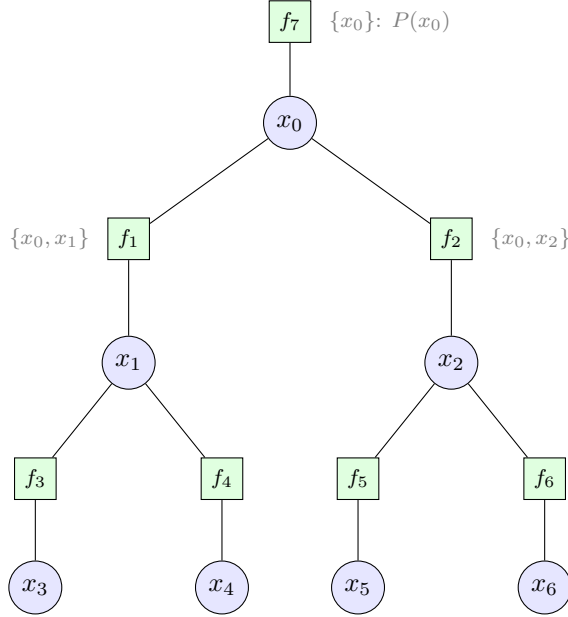


Figure 2: The bipartite *factor graph* of **tree.lcn**: circles are variable nodes (atoms), squares are factor nodes.  $f_7 = \{x_0\}$  carries the root prior; each  $f_i$  ( $i \leq 6$ ) carries the two transition sentences of one tree edge. Crucially the bipartite graph is a *tree* — removing any factor disconnects the graph, and any two variables communicate along a unique factor path. This is what makes the interval-message summary lossless (Section 5).

The chain’s factor graph is the special case where the tree is a path: factors  $f_i = \{x_{i-1}, x_i\}$  in a line plus the prior  $\{x_0\}$ . The polytree’s factor graph is also a tree, but with two 3-atom factors at the colliders:  $\{x_0, x_1, x_2\}, \{x_3, x_4, x_5\}$ , joined by  $\{x_2, x_3\}$  and  $\{x_5, x_6\}$  and the root priors  $\{x_0\}, \{x_1\}, \{x_4\}$ . In all three cases the factor graph is singly connected.

## 5 How the factor graph preserves the LMC

This is the crux. ARIEL never builds the global  $2^n$  joint, so it cannot impose the LMC equalities (1) globally. There are exactly two ways an LMC assertion can survive into ARIEL’s answer: it can be **imposed** (its full scope  $x \cup Y \cup S$  fits inside one factor, so (1) is added to that factor’s local NLP, step (ii) of Section 1), or it can be **preserved structurally** (the factor-graph topology and the message protocol already entail it). We examine which happens.

### 5.1 On the test structures, every LMC is dropped — yet exactness holds

Table 2 runs the contrast (`ArielInference.analyze`) for the three structures. The striking fact: *on all three, not one LMC assertion fits inside a single factor*, so *none* is imposed as an explicit constraint. The local NLPs contain only sentence rows and neighbour intervals — no LMC equalities at all.

| structure | #LMC assertions | #imposed (fit a factor) | why every LMC spans $\geq 2$ factors                                                                                              |
|-----------|-----------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| chain     | 6               | 0                       | e.g. $(x_2 \perp\!\!\!\perp x_0 \mid x_1)$ has scope $\{x_0, x_1, x_2\}$ ; factors are 2-atom edges, so $x_0, x_2$ never co-occur |
| tree      | 6               | 0                       | e.g. $(x_1 \perp\!\!\!\perp x_2, x_5, x_6 \mid x_0)$ spans the whole left/right subtrees                                          |
| polytree  | 7               | 0                       | e.g. $(x_0 \perp\!\!\!\perp x_1, x_4)$ has scope $\{x_0, x_1, x_4\}$ straddling <i>both</i> collider factors                      |

Table 2: On the chain, tree and polytree, *no* LMC assertion fits inside a single factor, so *none* is imposed as an explicit equality in any local NLP. Despite this, ARIEL is exact (Table 1). The LMC must therefore be preserved *structurally*.

So the LMC equalities are *not* enforced anywhere as constraints, yet the bounds are exact. The resolution is that on a singly-connected factor graph the message protocol *is* the LMC.

### 5.2 Why the interval summary is exactly the LMC factorization

Consider a factor  $f$  with target  $n$  and one other boundary variable  $m$  (the chain/tree case). When ARIEL computes  $f \rightarrow n$ , the variable  $m$  enters the local NLP only through its marginal interval. Equivalently,  $f$  treats  $m$  as *independent of everything reached through  $m$ 's other factors, given the value summarized by the interval*. On the tree of Figure 2,  $m$  is precisely a *separator*: deleting  $m$  from the factor tree disconnects  $f$ 's side from the rest. The variables on the far side of  $m$  are LMC-independent of  $n$  given  $m$  — that is exactly assertion (1) with  $S = \{m\}$ . Thus

*the interval message across a separator  $m$  encodes the LMC independence “everything on  $f$ 's side  $\perp\!\!\!\perp$  everything beyond  $m \mid m$ ” — the very assertion that was “dropped”.*

Because the factor graph is a tree, every LMC assertion decomposes into such single-separator statements along the unique path between the relevant variables, and each is realized by one interval message. No global joint, and no explicit equality, is needed: the topology supplies the factorization. This is the LCN analogue of why sum-product belief propagation is exact on trees.

### 5.3 The collider case: the LMC lives *inside* a factor's joint

The polytree is more subtle because moralizing a collider  $x_0, x_1 \rightarrow x_2$  puts all three atoms in one factor  $\{x_0, x_1, x_2\}$  (its sentences are  $P(x_2 \mid x_0, x_1)$ , which mention all three). The relevant LMC here is the *co-parent* independence  $(x_0 \perp\!\!\!\perp x_1)$ . Two observations make this exact:

1. **Inside the factor, the full  $2^3$  joint is modelled**, so the collider conditional  $P(x_2 \mid x_0, x_1)$  is represented exactly — nothing about the  $x_0$ - $x_1$ - $x_2$  interaction is approximated within  $f$ .
2. **The co-parent independence  $(x_0 \perp\!\!\!\perp x_1)$  is *not* imposed** (its scope  $\{x_0, x_1\}$  does fit the factor, but the assertion the LMC actually generates is  $(x_0 \perp\!\!\!\perp x_1, x_4)$ , scope  $\{x_0, x_1, x_4\}$ ,

which spans both colliders — Table 2). It does not need to be: the only messages *leaving* the collider factor go to  $x_2$  (toward the rest of the chain) and back to the priors  $x_0, x_1$ ; along the factor tree,  $x_0$  and  $x_1$  have no second path to influence each other, so leaving them free inside  $f$  cannot create spurious correlation in any marginal. The brute-force oracle confirms the resulting  $P(x_2), P(x_3), \dots$  are exact (Table 1).

Indeed ARIEL’s outgoing messages *implicitly assert* pairwise independencies among a factor’s boundary that are *not* in the LMC — for the collider factor these include the (false)  $(x_0 \perp\!\!\!\perp x_2)$  and  $(x_1 \perp\!\!\!\perp x_2)$  (**analyze** reports them under “added by ARIEL, not in LMC”). These are harmless on a tree precisely because they only ever summarize a message along a single factor path; there is no returning path on which the missing correlation could matter.

#### 5.4 When an LMC *does* fit a factor it *is* imposed

The structural mechanism above is not the whole story — step (ii) of Section 1 is live. When an LMC assertion’s full scope happens to fit one factor, ARIEL adds its equality (1) to that factor’s local NLP. For instance on **asia** the assertion  $(C \perp\!\!\!\perp B \mid S)$  has scope  $\{B, C, S\}$ , which equals the factor  $\{B, C, S\}$ , and **analyze** reports it as *enforced* (while the three wider asia assertions are dropped/structural). So ARIEL’s LMC handling is two-tier: **impose where the scope fits a factor; rely on the factor tree elsewhere**. On a singly-connected graph the second tier is exact; the first tier is a bonus that tightens factors with  $\geq 3$  atoms.

#### 5.5 Summary of LMC preservation

**Proposition 1** (LMC preservation on singly-connected factor graphs). *Let the factor graph  $\mathcal{F}$  be singly connected (a tree). Then for every LMC assertion  $(x \perp\!\!\!\perp Y \mid S)$ :*

- *if  $x \cup Y \cup S \subseteq \text{scope}(f)$  for some factor  $f$ , the equality (1) is imposed in  $f$ ’s local NLP (exact, by construction); otherwise*
- *the assertion factorizes into single-separator independencies along the unique factor path connecting  $x$ ,  $Y$  and  $S$ , each of which is realized exactly by an interval message across that separator.*

*In both cases the assertion holds in ARIEL’s converged solution, so the bounds equal the exact LCN bounds (Table 1). The interval summary is information-lossless precisely because a tree has no second path along which a discarded joint correlation could matter.*

## 6 The boundary: a loopy instance where the LMC is no longer preserved

Proposition 1 fails the moment the factor graph has a cycle, because then a separator no longer disconnects the graph and the interval summary drops correlation that a *second* path needs. We exhibit this with the **diamond**  $A \rightarrow \{B, C\}, \{B, C\} \rightarrow D$  (collider at  $D$ ) plus a single *cross-family* sentence coupling the co-parents:

$$P(A) \in [0.4, 0.6], \quad P(B \mid A), P(C \mid A), \quad P(D \mid B, C) \text{ (4 rows),} \quad \boxed{P(B \wedge C) \leq 0.25}.$$

The skeleton has the 4-cycle  $A-B-D-C-A$  and the extra sentence  $P(B \wedge C)$  adds a second loop, so the factor graph is *loopy* (2 independent cycles); the instance is consistent. Table 3 compares

ARIEL with the exact bounds (confirmed by both **CredalJT** and the brute-force oracle, LMC residual  $2 \times 10^{-16}$ ).

| atom     | ARIEL                    | exact                    | verdict              |
|----------|--------------------------|--------------------------|----------------------|
| <i>A</i> | [0.4000, 0.6000]         | [0.4000, 0.6000]         | exact                |
| <i>B</i> | [0.3600, <b>0.6400</b> ] | [0.3600, <b>0.5700</b> ] | <i>outer</i> (+0.07) |
| <i>C</i> | [0.3600, <b>0.6400</b> ] | [0.3600, <b>0.5700</b> ] | <i>outer</i> (+0.07) |
| <i>D</i> | [0.3160, <b>0.6840</b> ] | [0.3160, <b>0.5940</b> ] | <i>outer</i> (+0.09) |

Table 3: On the loopy coupled diamond, ARIEL is only a valid *outer* bound: the upper bounds of *B*, *C*, *D* are 0.07–0.09 too high. The cross-family sentence  $P(B \wedge C) \leq 0.25$  lives in the factor  $\{B, C\}$ , but the messages it sends to *B* and to *C* carry only the *marginal* intervals  $P(B)$ ,  $P(C)$  — they cannot transmit the *joint* restriction on  $B \wedge C$  around the loop, so the cap never tightens the rest.

This is the interval-message approximation of Section 1.5 biting: the factor  $\{B, C\}$  *knows*  $P(B \wedge C) \leq 0.25$  but can only *report* marginal intervals on *B* and *C* to the rest of the graph. The joint information is lost at the message boundary — exactly the failure mode loopy belief propagation exhibits in classical graphical models. The same mechanism would make a *spanning* LMC equality non-preservable on a loopy graph: with two paths between two variables, the single-separator decomposition of Proposition 1 no longer exists. ARIEL remains *sound* (its interval contains the truth) but not *tight*.

## 7 Can ARIEL ever be exact? — the precise answer

**Proposition 2** (What the experiments establish). *For the marginal task on binary LCNs:*

- (i) **Yes, ARIEL is exact on singly-connected LCNs.** *It returns the exact posterior marginal bounds on the **chain**, the branching **tree**, and the **polytree** (Table 1), confirming Theorem 1. It is also exact on several loopy instances whose cycles do not bind (e.g. **cancer**, **asia**, **d4 biting**, the uncoupled diamond), but that is incidental, not guaranteed.*
- (ii) **Not in general on loopy LCNs.** *When a cross-family constraint (or a spanning LMC) couples variables that the factor graph can only separate through marginal intervals (the coupled diamond, Table 3), ARIEL is a valid outer bound but not tight. This is inherent to interval message passing and matches the paper’s restriction of Theorem 1 to the singly-connected case.*
- (iii) **The LMC is preserved without ever being globally imposed.** *On the test structures no LMC fits a single factor (Table 2); the LMC survives structurally, because the bipartite factor graph is a tree and each (dropped) assertion decomposes into single-separator independencies realized exactly by the interval messages (Proposition 1). When a scope does fit a factor (e.g. **asia**’s  $(C \perp\!\!\!\perp B \mid S)$ ), the equality is additionally imposed inside that factor’s local NLP.*

**Practical takeaway.** ARIEL’s exactness on chains, trees and polytrees is real and rests on a clean principle: *a singly-connected factor graph turns every LMC independence into a sequence of separator marginals that interval messages transmit losslessly.* The exact moment that principle breaks is a cycle in the factor graph — introduced by a cross-family sentence or by a moralization that leaves two variables with a second communication path — after which ARIEL is a sound outer bound rather than an exact one.



**Reproducibility.** All numbers come from `examples/chain.lcn`, `examples/tree.lcn`, `examples/polytree.lcn` and the coupled diamond ( $P(B \wedge C) \leq 0.25$  added to the four-atom collider). The factor graphs and the LMC contrast are produced by `ArielInference.analyze`. The exact references are `CredalJT (cn/junction_nlp.py, solver="scip")` and an LMC-correct parametric brute-force oracle (SLSQP).